

GPU Confidential Computing

Threat Model Analysis: Non-CC GPU vs Bounce-Buffer CC vs TDISP

June 2026 · For internal infrastructure planning

This report provides a structured threat model analysis of NVIDIA GPU confidential computing across three architectural configurations: SEV-SNP with a non-CC GPU (T4, L4, A2, RTX — no GPU TEE), bounce-buffer CC (H100/H200 with GPU TEE, available today), and TDISP/TEE-I/O inline encryption (Blackwell, not yet GA). It maps each threat actor and attack vector to the capabilities of each configuration, and provides guidance on which regulatory and contractual requirements each satisfies.

1. Technical Primer: What Each Mode Actually Does

1.1 The problem both modes solve

A standard (non-confidential) GPU deployment has no isolation guarantees between the host OS/hypervisor and the GPU's computation. The hypervisor can read any data in flight to or from the GPU. DMA buffers, model weights loaded into VRAM, activations, KV cache, and output tokens are all visible to a sufficiently privileged attacker on the host.

Both CC modes address this by establishing cryptographic isolation between a Confidential VM (CVM) running the inference workload and the untrusted host stack. The key difference is *where* the cryptographic boundary sits and whether the PCIe bus between CPU and GPU is inside or outside that boundary.

1.2 Bounce-buffer CC (Hopper: H100, H200 — shipping today)

In bounce-buffer mode, the CPU CVM and the GPU maintain *separate* TEEs that verify each other through attestation but do not share a unified encrypted domain. Data movement between them works as follows:

- The CVM encrypts data in its own protected memory (using SEV-SNP page encryption keys managed by the AMD Secure Processor).
- To transfer data to the GPU, the driver copies it to a shared non-confidential bounce buffer page — a region of memory accessible to both the CVM and the GPU's DMA engine.
- The GPU DMA engine reads from the bounce buffer and moves data into its own protected VRAM, which is encrypted by the GPU's internal key (separate from the CPU's SEV-SNP keys).
- The reverse path applies on return: GPU decrypts from VRAM, writes to bounce buffer, CVM reads and re-encrypts into its protected memory.

■ The bounce buffer itself is not plaintext in DRAM — it is encrypted at the CPU's memory-encryption layer (via AMD SME for the page, even if outside the CVM's private address space). However, it is **not** protected by the CVM's VM-specific key, meaning the host hypervisor's key hierarchy can theoretically access it. Whether this constitutes a real attack depends on your threat model and is the central question this report examines.

The attestation flow in bounce-buffer CC: the CVM requests a GPU attestation report via the NVIDIA driver inside the CVM. The GPU's on-die Secure Processor signs a report covering the GPU's firmware, configuration, and CC mode status. The CVM verifies this report against NVIDIA's Root CA before releasing any sensitive data (model weights, user inputs). From the tenant's perspective, they have verified: (1) the GPU is a genuine NVIDIA H100/H200, (2) CC mode is enabled, (3) the GPU firmware is unmodified. They have *not* verified the integrity of the PCIe bus between CPU and GPU.

1.3 TDISP + PCIe IDE (Blackwell — not GA as of mid-2026)

TDISP (TEE Device Interface Security Protocol, PCI-SIG PCIe r6.0) combined with PCIe IDE (Integrity and Data Encryption) eliminates the architectural separation between the CPU TEE and the GPU. Instead of two separate TEEs that attest to each other, the GPU becomes a hardware-verified *extension* of the CVM's trusted domain.

- PCIe IDE provides AES-256-GCM encryption on the PCIe link at the physical layer, between the CPU's PCIe root complex and the GPU's endpoint — before data touches any DRAM buffer.
- TDISP provides the protocol by which the CVM binds the GPU device to its TEE: the CVM issues TDISP commands (Interface Report, Lock, etc.) that are verified by the GPU's Secure Processor and recorded in a device interface report that extends the attestation chain.
- The result is a single unified attestation report covering both the CPU execution environment and the attached GPU device — the tenant verifies one chain, not two.
- No bounce buffer is needed: the GPU's DMA engine reads and writes directly into the CVM's private encrypted address space over the IDE-protected PCIe link.

■ TDISP is not GA as of mid-2026. AMD SEV-TIO (the AMD implementation of TDISP for EPYC Turin) landed only its first upstream Linux phase in the 6.19 merge window (December 2025). NVIDIA's R590 GA driver ships Blackwell CC only in bounce-buffer modes. Treat TDISP as a 2027 build target, not a 2026 deployment option.

2. Threat Actor Taxonomy

Before mapping attack vectors to CC modes, it is necessary to be precise about which adversaries are in scope. GPU confidential computing protects against a specific and limited set of threats — not all of them. The following taxonomy structures adversaries by access level and capability.

2.1 Tier 1 — Software-privileged host adversary

The primary adversary both CC modes are designed to defeat. This actor has root or hypervisor-level access to the host OS and can read arbitrary DRAM, manipulate DMA mappings, and inspect kernel state — but cannot physically probe hardware or bypass CPU/GPU silicon-level protections.

- Examples: a malicious hypervisor operator, a compromised host OS, a rogue sysadmin with root access, a cloud provider employee with hypervisor access.
- Capability: can read all unprotected DRAM, manipulate page tables, inspect DMA buffers, and replay or modify I/O that crosses the software layer.
- Cannot: break AES-256 encryption, forge signed attestation reports, read CPU SEV-SNP-encrypted memory, or physically tap the PCIe bus.

2.2 Tier 2 — Physical-access datacenter insider

An actor with physical access to the server — the key adversary that separates bounce-buffer CC from TDISP. This actor can install hardware interposers, probe PCIe signals with protocol analyzers, or observe electrical signals on the motherboard. They may or may not also have software-level access.

- Examples: a data-center technician with physical access to an unlocked rack, a nation-state actor with supply-chain access, a malicious colocation facility employee.
- Capability: can tap the PCIe bus between CPU and GPU, observe data in transit over the PCIe link if it is not encrypted at the physical layer, potentially modify hardware.
- Cannot (without additional capability): break AES-256-GCM on an IDE-protected link, forge SPDMM session keys, access GPU's internal protected VRAM without the GPU's key.

2.3 Tier 3 — Supply-chain / firmware adversary

An actor who has compromised hardware or firmware before deployment — implanting malicious firmware in the GPU, the BMC, the PCIe switch, or the CPU microcode. Both CC modes address this threat through measured boot and attestation; neither protects against a hardware-level backdoor implanted in the GPU's Secure Processor itself.

- Examples: a compromised manufacturing process, a nation-state firmware implant, a rogue OEM firmware update.
- Capability: can subvert the attestation root of trust if the compromise occurs at or below the Root CA level.
- Mitigation: NVIDIA's attestation chain roots to an on-die fused key; firmware updates are signed and measured. The threat is real but requires very sophisticated access.

2.4 Tier 4 — Network / software adversary (not a GPU CC concern)

Network attackers, application-layer exploits, and software vulnerabilities in the model serving stack are outside the scope of GPU confidential computing. CC does not protect against a compromised vLLM process, a prompt injection attack, or a network attacker who intercepts inference requests before they enter the TEE. These require separate controls (TLS termination inside the TEE, input validation, rate limiting).

■ GPU CC is not a substitute for application security. It protects **data in use inside the GPU**. Data in transit to/from the inference endpoint requires separate TLS/mTLS controls. Data at rest (model weights on NVMe) requires disk encryption. CC addresses only the runtime execution environment.

3. Attack Vector Analysis

The following sections analyse each significant attack vector, map it to a threat actor tier, and assess the effectiveness of three configurations: SEV-SNP with a non-CC GPU (no GPU TEE), bounce-buffer CC (H100/H200), and TDISP+IDE (Blackwell).

3.1 Hypervisor reads CPU memory

A hypervisor-level attacker attempts to read the CVM's CPU-side memory — model orchestration state, API keys, tenant credentials, or data staged for GPU transfer.

- SEV-SNP + non-CC GPU: DEFEATED. SEV-SNP protects CPU memory regardless of GPU type.
- Bounce-buffer CC: DEFEATED. Same SEV-SNP protection, plus GPU VRAM is separately encrypted.
- TDISP: DEFEATED (same mechanism plus IDE link encryption).

3.2 Hypervisor reads GPU VRAM

A hypervisor-level attacker attempts to read data resident in GPU VRAM — model weights, activations, KV cache, and output tokens stored on the GPU during inference.

- SEV-SNP + non-CC GPU: NOT MITIGATED. Non-CC GPUs (T4, L4, A2, RTX) have no VRAM encryption. The host can read all GPU memory directly. Model weights loaded to the GPU, inference activations, KV cache, and output tokens are fully exposed.
- Bounce-buffer CC: DEFEATED. The GPU's internal key encrypts VRAM; the host cannot read it.
- TDISP: DEFEATED (same GPU-side encryption).

✗ This is the most significant gap in the non-CC GPU configuration. If model weights or inference inputs/outputs are confidential, any GPU without CC mode is a full exposure of that data to the host — regardless of CPU-side SEV-SNP protection.

3.3 DMA remapping attack

The hypervisor attempts to remap DMA addresses to redirect GPU DMA reads/writes to attacker-controlled memory regions, or to cause the GPU to DMA sensitive data to a region the hypervisor can read.

- SEV-SNP + non-CC GPU: PARTIAL. SNP protects CPU-side page mappings, but GPU DMA targets are unprotected — the hypervisor can redirect GPU DMA to observe or tamper with data in transit.
- Bounce-buffer CC: DEFEATED. SEV-SNP's Secure Nested Paging (the SNP extension) prevents the hypervisor from remapping guest physical addresses without the guest's knowledge. Memory integrity protection validates that pages have not been remapped or replayed.
- TDISP: DEFEATED (same SNP protection plus TDISP locks the GPU device interface to the CVM, preventing the hypervisor from redirecting GPU DMA).

■ This is why EPYC Milan (7003) or newer is a hard requirement — not just for memory encryption (SEV), but for memory *integrity* (SNP). A Rome (7002) host with SEV but no SNP is vulnerable to DMA remapping attacks even with GPU CC mode enabled.

3.4 Bounce-buffer / CPU-GPU DMA observation

The hypervisor attempts to read data as it transits between CPU and GPU — either via the bounce buffer (in CC mode) or via standard DMA (in non-CC mode).

In non-CC configurations, this is trivial: CPU-to-GPU DMA is plaintext and fully observable by the host. In bounce-buffer CC, the staging area is encrypted at the platform level (SME) but not by the CVM's private key.

- SEV-SNP + non-CC GPU: NOT MITIGATED. DMA between CPU and GPU is plaintext; no bounce buffer, no encryption on the transfer path.
- Bounce-buffer CC: PARTIALLY MITIGATED. The bounce buffer is encrypted at the platform level (SME), so a software adversary reading raw DRAM sees ciphertext. However, a hypervisor with access to the platform memory-encryption key (which it may hold, depending on the platform configuration) could theoretically decrypt the bounce buffer. The practical exploitability of this depends on whether the hypervisor can access the SME key — on a well-configured SEV-SNP deployment, the hypervisor does not hold the VM-private key but may hold the platform key used for non-VM memory.
- TDISP: ELIMINATED. No bounce buffer exists. Data moves directly from the CVM's private encrypted memory over the IDE-protected PCIe link into the GPU's protected VRAM. There is no shared staging area.

■ For most realistic software-adversary threat models, the bounce buffer is not a practical vulnerability — the hypervisor's access to the platform SME key is constrained by firmware, and exploiting this path requires both hypervisor compromise AND the ability to use the platform key. However, the architectural gap is real and is the reason TDISP matters for high-assurance deployments.

3.5 PCIe bus physical tap

A physically-present attacker installs a PCIe protocol analyzer or interposer between the CPU's root complex and the GPU endpoint, and captures the raw PCIe traffic to reconstruct data in transit.

- SEV-SNP + non-CC GPU: NOT MITIGATED. The PCIe bus is unencrypted; additionally, since the GPU has no VRAM encryption, a physical tap exposes both the transfer and the destination.
- Bounce-buffer CC: NOT MITIGATED. The PCIe bus is not encrypted at the physical layer in bounce-buffer mode. A sufficiently capable hardware attacker with physical access to the server's PCIe slot or riser card can capture DMA traffic between the CPU and GPU. PCIe Gen4/Gen5 operates at 16–32 GT/s per lane; x16 slots require specialized (expensive) equipment to tap at full bandwidth, but the attack is technically feasible.
- TDISP: DEFEATED. PCIe IDE encrypts all TLP (Transaction Layer Packet) data with AES-256-GCM at the physical layer. An interposer on the PCIe bus sees only ciphertext; the session keys are negotiated via SPDH between the CPU root complex and the GPU endpoint and are not accessible to an external observer.

■ PCIe bus tapping is expensive and requires physical access, making it a Tier 2 threat. For most enterprise and cloud deployments, physical access controls (locked racks, CCTV, two-person access rules) are sufficient to downgrade this threat — TDISP's protection becomes necessary only when those physical controls are not trusted or cannot be verified by the tenant.

3.6 NVLink multi-GPU observation

In a multi-GPU configuration, an attacker attempts to observe GPU-to-GPU traffic carried over NVLink — tensor-parallel activations, all-reduce gradients, or KV cache shards transferred between GPUs.

- SEV-SNP + non-CC GPU: NOT MITIGATED. NVLink traffic on non-CC GPUs is plaintext with no hardware isolation between GPU instances.
- Bounce-buffer CC (Hopper single-GPU SPT): NOT APPLICABLE. Single-GPU CC does not involve NVLink.
- Bounce-buffer CC (Hopper PCIe multi-GPU): NOT MITIGATED. NVLink GPU-to-GPU traffic in Hopper PCIe mode is plaintext. A physically-present attacker with access to the HGX backplane can observe inter-GPU traffic. This is a significant limitation for multi-GPU inference workloads requiring confidentiality.
- TDISP (Blackwell MPT CC): DEFEATED. Blackwell's secure multi-GPU domain encrypts NVLink traffic within the TEE boundary. GPU-to-GPU communication is protected by inline link-layer encryption.

■ For single-GPU inference (models that fit on one H100/H200 80–94GB), the NVLink gap is irrelevant. For models requiring tensor parallelism across multiple GPUs (70B+ in FP16, 400B+ in FP8), the Hopper PCIe gap is material and is a genuine argument for waiting for Blackwell MPT CC.

3.7 Attestation

Can the tenant verify the integrity of the execution environment?

- SEV-SNP + non-CC GPU: PARTIAL. CPU attestation (SEV-SNP) works — the tenant can verify the CVM is running on genuine hardware with unmodified firmware. GPU attestation is NOT AVAILABLE — non-CC GPUs have no on-die Secure Processor, no attestation chain, and no way for the tenant to verify the GPU's identity or configuration.
- Both CC modes: GPU attestation DEFEATED by NVIDIA's attestation chain. The GPU's on-die RIM (Reference Integrity Manifest) is fused at manufacturing; attestation reports are signed by the GPU's Secure Processor using a key that chains to NVIDIA's Root CA. Forging a report requires breaking the GPU's fused key or compromising NVIDIA's Root CA — neither is tractable.
- Caveat: attestation verification must be implemented correctly by the tenant. A misconfigured verifier that does not check the certificate chain, does not validate the report's freshness nonce, or does not enforce CC-mode-on status silently negates this protection. Treat attestation logic as security-critical code.

3.8 Firmware implant (supply-chain)

A compromised firmware update process installs malicious GPU firmware that exfiltrates data from VRAM or subverts the attestation report.

- SEV-SNP + non-CC GPU: NOT APPLICABLE. Without GPU attestation, firmware integrity is unverifiable by the tenant — a firmware implant is undetectable, but also unnecessary since VRAM is already readable by the host without one.
- Both CC modes: PARTIALLY MITIGATED. NVIDIA's firmware update process requires signed updates that chain to the on-die key; an unsigned or differently-signed firmware image will not load. However, if the attacker has compromised NVIDIA's signing infrastructure or the OEM's firmware customization process, they can produce a signed-but-malicious firmware image that passes attestation.

- Mitigation: NVIDIA provides Reference Integrity Manifests (RIMs) covering expected firmware measurements; a tenant's attestation verifier should check that the measured firmware matches a known-good RIM, not just that the attestation signature is valid.

4. Threat Coverage Matrix

The following table summarises which threats each CC mode defeats, partially mitigates, or does not address. 'Practical' refers to the realistic exploitability given normal datacenter physical controls; 'Theoretical' refers to architectural gaps that are not practically exploitable in most deployments but are present at the protocol level.

Attack vector	Tier	SEV-SNP + non-CC GPU	Bounce-buffer CC (H100/H200)	TDISP+IDE (Blackwell)	Notes
Hypervisor reads CPU memory	T1	DEFEATED	DEFEATED	DEFEATED	SEV-SNP protects CPU memory in all configs
Hypervisor reads GPU VRAM	T1	NOT MITIGATED	DEFEATED	DEFEATED	Non-CC GPUs have no VRAM encryption
CPU-GPU DMA observation	T1	NOT MITIGATED	Theoretical gap	ELIMINATED	Non-CC: plaintext DMA. BB: SME-encrypted staging
DMA remapping	T1	PARTIAL	DEFEATED	DEFEATED	SNP protects CPU pages; GPU side unprotected w/o CC
GPU-side model weight extraction	T1	NOT MITIGATED	DEFEATED	DEFEATED	Weights in GPU memory exposed without CC
GPU-side KV cache / activation snoop	T1	NOT MITIGATED	DEFEATED	DEFEATED	Inference I/O in GPU memory exposed without CC
PCIe bus physical tap	T2	NOT MITIGATED	NOT MITIGATED	DEFEATED	No config encrypts PCIe physically except TDISP/IDE
NVLink multi-GPU observation	T2	NOT MITIGATED	NOT MITIGATED	DEFEATED	Hopper PCIe: plaintext NVLink. Blackwell: encrypted
GPU attestation	T3	NOT AVAILABLE	DEFEATED	DEFEATED	Non-CC GPUs have no attestation capability
CPU attestation (SEV-SNP)	T3	DEFEATED	DEFEATED	DEFEATED	CPU attestation works regardless of GPU type
Firmware implant (signed)	T3	N/A	Partial (RIM check)	Partial (RIM check)	Non-CC: unverifiable. CC: check RIMs
Application exploit	T4	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE	Requires app-layer controls
Network interception	T4	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE	Requires TLS inside TEE

T1 = Software-privileged host adversary. T2 = Physical-access datacenter insider. T3 = Supply-chain/firmware adversary. T4 = Network/software adversary (out of GPU CC scope).

■ **Key takeaway:** SEV-SNP + non-CC GPU protects everything in CPU memory (orchestration, credentials, model loading logic) but offers **zero protection** for anything that touches the GPU — VRAM, DMA transfers, model weights, activations, and KV cache are all visible to the host. This is acceptable

when the GPU processes only non-sensitive data. It is NOT acceptable when model weights or inference inputs/outputs are confidential.

5. Regulatory and Contractual Requirement Mapping

Different compliance frameworks and contractual requirements impose different standards for 'data in use' protection. The following analysis maps the most common frameworks to the appropriate CC mode.

5.1 Frameworks satisfied by bounce-buffer CC alone

The following frameworks' requirements for data-in-use protection are met by bounce-buffer CC on H100/H200, assuming correct deployment and attestation:

- SOC 2 Type II (confidentiality and security criteria): bounce-buffer CC satisfies the requirement that data in use be protected from privileged operator access. The attestation report provides auditable evidence.
- HIPAA (45 CFR §164.312(a)(2)(iv), encryption): HHS has not mandated a specific encryption standard for data in use; demonstrable encryption of PHI during GPU processing via CC attestation satisfies the 'addressable' encryption specification.
- ISO 27001 (A.10.1 Cryptographic controls): CC mode with attestation satisfies data-in-use encryption controls for personal data processed by GPU inference workloads.
- GDPR Article 32 (appropriate technical measures): The European Data Protection Board's guidance treats hardware TEEs with attestation as appropriate technical measures for data-in-use protection. Bounce-buffer CC with proper attestation verification satisfies this in most member-state interpretations.
- FIPS 140-2/140-3: NVIDIA H100/H200 use AES-256-GCM for VRAM encryption; the attestation chain uses standard NIST algorithms. FIPS compliance depends on the overall system stack, not GPU CC alone, but CC does not introduce non-FIPS primitives.

5.2 Frameworks likely requiring TDISP (or equivalent physical controls)

The following frameworks impose requirements that bounce-buffer CC cannot satisfy without supplementary physical controls, or that TDISP satisfies more cleanly:

- Common Criteria EAL 5+ (high assurance): CC EAL 5 requires formal security architecture analysis and resistance to 'high' attack potential. The bounce-buffer architectural gap — even if not practically exploitable — may not survive formal analysis without TDISP, depending on the evaluated configuration.
- NIST SP 800-207 (Zero Trust Architecture): ZTA requires that 'all data sources and computing services are considered resources' and that trust is established dynamically per-request. A unified attestation report (TDISP) covering both the CPU and GPU as a single trusted resource is architecturally cleaner than two separate attestation reports. Bounce-buffer CC can satisfy ZTA principles but requires careful attestation composition logic.
- DoD IL5/IL6 (high-sensitivity defense workloads): DoD CIO guidance for classified and CUI processing generally requires protection against insider threats including physical access. IL6 (classified) workloads may require TDISP-equivalent physical link encryption, depending on the specific overlay and accreditor interpretation.
- PCI DSS v4.0 (cardholder data in AI inference): PCI DSS Requirement 3.5 mandates encryption of stored cardholder data; for in-use data during inference, the PCIe-bus gap in bounce-buffer CC may be flagged in a QSA assessment if the hosting facility does not provide independently verified physical access controls.

5.3 Multi-tenant inference: the additional dimension

For shared/multi-tenant LLM inference — where multiple customers' requests are processed by the same GPU fleet — there is an additional dimension beyond what a single-tenant threat model captures.

In single-tenant CC, the question is: can the host observe the tenant's data? In multi-tenant CC, the additional questions are: can one tenant observe another tenant's data? Can the model provider observe a tenant's inputs? Can a tenant extract the model provider's proprietary weights?

- MIG (Multi-Instance GPU) provides hardware-level memory partitioning between GPU instances but does not encrypt inter-instance traffic or provide per-tenant attestation. MIG + CC-mode gives one CVM the whole GPU; MIG slices within that CVM are tenant-separated by hardware memory bounds, not encryption.
- For strong multi-tenant isolation with attestation, the clean architecture is one CC-mode GPU per tenant CVM — matching physical isolation to cryptographic isolation. This is expensive but removes the intra-GPU sharing ambiguity.
- Model weight protection (from tenant to model provider) and input protection (from model provider to tenant) are dual requirements that pull in opposite directions. Both require the CVM to be the trust anchor, with weights and inputs each encrypted to keys the other party doesn't hold. This is achievable with bounce-buffer CC via careful key management but requires explicit design.

■ The SaaS multi-tenant inference threat model is meaningfully more complex than the enterprise private-deployment model. If your deployment serves multiple independent tenants on shared GPU hardware, engage a security architect to design the specific key management, attestation composition, and isolation architecture before deploying.

6. Decision Framework

The following questions determine which CC mode is appropriate for a given deployment. Work through them in order.

6.1 Is physical access to the server trusted?

If yes (locked colo cage with independent physical access controls, CCTV with audit log, two-person access rules, no PCIe slot exposure) — bounce-buffer CC is architecturally sufficient for the PCIe-tap threat. The physical controls substitute for PCIe IDE.

If no (shared facility, uncertain physical controls, tenant has no visibility into physical access logs) — TDISP is required to cryptographically remove the physical-access attack vector. Since TDISP is not GA, the current options are: (a) negotiate facility-level physical controls and document them for compliance purposes, or (b) defer GPU confidential inference until TDISP is available.

6.2 Does the workload require multi-GPU tensor parallelism?

If no (models fit on a single H100/H200 80–94GB, or quantized 70B models fit on H100 NVL 94GB in FP8) — single-GPU SPT CC mode is appropriate. NVLink is not involved and Hopper is sufficient.

If yes (full-precision 70B+ or 400B+ quantized requiring multi-GPU all-reduce) — Hopper PCIe CC has an unencrypted NVLink gap. Options: (a) use Blackwell MPT CC (encrypted NVLink, bounce-buffer for CPU↔GPU), or (b) determine whether the NVLink gap is in scope for your threat model (it requires physical access to the HGX backplane).

6.3 Does your compliance framework require a unified attestation report?

Some high-assurance frameworks (DoD IL6, CC EAL 5+, certain FedRAMP High overlays) may require a single attestation report that covers the entire trusted execution boundary without compositional ambiguity. Bounce-buffer CC produces two separate attestation reports (CPU CVM + GPU) that must be composed by the verifier. TDISP produces one.

If a unified report is required — wait for TDISP, or use a cloud provider (Azure NCCads_H100_v5) where the composed attestation is handled by the platform's attestation service with a single tenant-facing attestation API.

6.4 Is the deployment self-managed on bare metal?

The bare-metal same-node SEV-SNP + GPU CC stack (as opposed to a managed cloud offering) requires careful version management of kernel, QEMU, OVMF, driver, and firmware. Known compatibility issues exist at multiple layers. For a production self-managed deployment:

- Separate-pool architecture (GPU pool separate from CPU application nodes) is substantially more stable and operationally mature than same-node passthrough.
- The separate pool requires an attestation bridge: the CPU application VM must establish an attested TLS channel to the GPU CVM before releasing sensitive data.

- Use QEMU 9.2.1 (not 10.1 under KubeVirt), OVMF.amdsev.fd, kernel ≥ 6.11 , and NVIDIA driver R590 GA. Pin these versions until you have validated a new combination end-to-end.

6.5 Summary decision table

Question	Answer	Architecture	Mode
GPU processes confidential data?	No	CPU fleet with non-CC GPU	SEV-SNP only — no GPU CC needed
GPU processes confidential data?	Yes	CC-mode GPU pool	Bounce-buffer CC or TDISP (see below)
Physical access trusted?	Yes	Same-node or separate pool	Bounce-buffer CC (H100/H200)
Physical access trusted?	No	Separate pool	Wait for TDISP (Blackwell + Turin, 2027)
Multi-GPU tensor parallel?	No	Separate pool recommended	SPT CC, single H100/H200 per CVM
Multi-GPU tensor parallel?	Yes (≤ 8 GPU)	HGX/DGX node	Blackwell MPT CC (NVLink encrypted)
Unified attestation report?	No	Either	Bounce-buffer CC sufficient
Unified attestation report?	Yes	Managed or TDISP	Azure NCCads_H100_v5 today; TDISP when GA
Self-managed bare metal?	Yes	Separate pool	Bounce-buffer CC (avoid same-node in prod)
SaaS multi-tenant?	Yes	One GPU per tenant CVM	Bounce-buffer CC with explicit key mgmt

7. Forward-Looking: TDISP Readiness Timeline

Given TDISP's importance for high-assurance deployments, the following milestones should be tracked to determine when bare-metal TDISP becomes viable.

Milestone	Status (mid-2026)	Target
PCIe IDE link encryption in Linux kernel (AMD SEV-TIO phase 1)	Merged — Linux 6.19 (Dec 2025)	Complete
Full TDISP bind/guest_request upstream (SEV-TIO end-to-end)	RFC/in-progress (Linux 6.20/7.0 window)	Late 2026
NVIDIA R-branch GA driver with TDISP inline mode	Not yet (R590 = bounce-buffer only)	2027 est.
EPYC Turin + Blackwell validated platform (OEM certified)	No shipping certified combo	2027 est.
TDISP-capable PCIe switch (if needed for multi-slot)	Early sampling (Microchip Switchtec)	2027 est.
Cloud provider TDISP offering (comparable to Azure NCCads_H100_v5)	None announced	2027-2028 est.

The practical guidance: if a deployment decision must be made today for a workload that ultimately requires TDISP, the options are:

- Deploy bounce-buffer CC now with physical access controls substituting for PCIe IDE, and plan a TDISP migration on the 2027-2028 hardware refresh cycle.
- Use Azure NCCads_H100_v5 (GA SEV-SNP + H100 NVL, managed attestation) for workloads requiring the strongest available confidential GPU posture today, accepting the cloud cost premium (~\$8.90/GPU-hr).
- For workloads where the NVLink multi-GPU gap matters, use Blackwell MPT CC (encrypted NVLink, bounce-buffer for CPU-GPU) rather than Hopper PPCle — this eliminates the NVLink gap while waiting for TDISP to address the CPU-GPU link.

This report reflects the state of NVIDIA GPU confidential computing as of June 2026. TDISP/TEE-I/O is an active area of kernel and driver development; the timeline estimates above should be re-evaluated quarterly. Key sources: NVIDIA Trusted Computing Solutions Release Notes R590 (Dec 2025); AMD SEV-TIO Linux 6.19 merge; NVIDIA Secure AI with Blackwell and Hopper Whitepaper WP-12554-001 v1.3 (Aug 2025); PCI-SIG PCIe r6.0 TDISP/IDE specs.